Thermionic emission depending on surface characteristics and review of Stefan-Bolzmann law

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A thermopile was used to check the material dependency of emitted thermal radiation of a Leslie cube and a black body. The four sides of the Leslie cube were black matt, polished, white matt and grey matt. The matt surfaces emitted more radiant flux than the polished one. This in accordance to Kirchhoff's law of thermal radiation. The Stefan-Boltzmann law describes the obtained data not completly correct as the Leslie cube is a grey body and not a black body. To realize a black body as close as possible, a hollow cylinder inside an oven was observed. Here the Stefan-Boltzmann law holds. With the same setup it was confirmed that the black body also obeys Lambert's cosine law. Using the Leslie cube the transmissivity for infrared radiation of different filters were determined. The results are $(47.29 \pm 0.78)\%$ for a 0.5 mm thick p-doped silicon plate, $(77.50 \pm 0.78)\%$ for a 5 mm thick NaCl crystal, $(2.34 \pm 0.77)\%$ for a infrared filter and $(3.71 \pm 0.77)\%$ for 3 mm thick window glass.

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1 Introduction

In the beginnings of the 19nth century thermionic emission set off one of the the most important developments in modern physics. When in December Max Planck found Rayleigh's Law and Wien's Law incompatible in describing black body radiation, he came up with what was later known as Planck's Law laying the foundations for the concept of quantum mechanics in modern physics. From this law the earlier observed Stefan-Boltzmann law can be derived. It connects the radiant flux emitted from a black body to the temperature of the body and its surface. Nowadays, latter findings proof to be highly relevant in various contexts of modern society. For everyday life, heat signatures are used to detect cable ruptures or improve isolation of buildings. In defence, they enable self guiding systems as for example used in the fire and forget technology, when missiles contain e.g. heat seeking heads. To get a better understanding of this the thermal radiation of different sides of a Leslie cube is measured and the infrared permeability of different filter type are discussed. Finally, the Lambert Law giving the angle distribution of thermal radiation is confirmed.

2 Experiment

Since all matter with non zero temperature emits radiation it is of interest to analyse it's radiation behaviour. For thermal radiation this can be achieved by thermocouples that convert a temperature difference into a voltage caused by the Seebeck effect. One side of the thermocouple is oriented towards the radiation source and the other one away from it. The averted side stays at a constant comparison temperature while the exposed one is heated up by the radiation of the emitting source. This generates a voltage that can be measured. Due to it's very low output many thermocouples are connected in series to

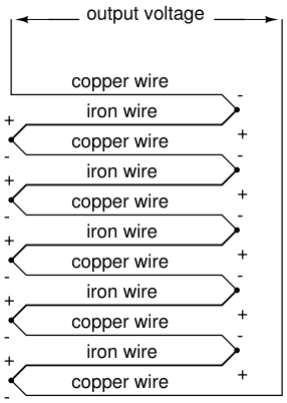

Fig. 1: Example of a thermopile, consisting of thermocouples made out of iron and copper. The dots represent junctions. One side is kept at a comparison temperature while the other is directed at the radiation source that's to be measured.

increase the voltage. This setup is called a thermopile and is exemplary depicted in Figure 1 for iron and copper wiring.

2.1 Temperature dependence of radiant power emitted by a Leslie cube

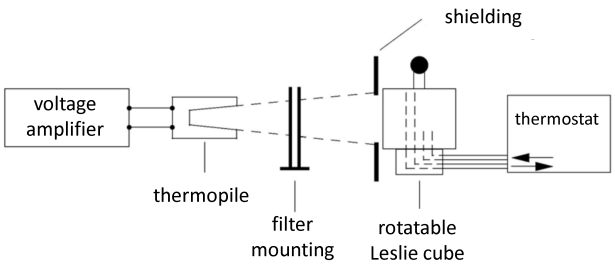

Fig. 2: Experimental setup fro the measurements with the Leslie cube [2] (edited). The cube emits radiation which is caught be the thermopile and converted to a voltage proportional to the radiation power. The shielding ensures that only the radiation of the Leslie cube is detected. The filter mounting is later used to determine the infrared transmissivity of several filters.

To observe the thermionic emission of a surface depending on it's surface properties a Leslie cube is used. The cube consists of four measured sites, black matt, white matt, nickel-plated and polished and grey matt. With a thermostat and water cooling/heating the temperature of the cube was set between values of 20 °C and 85 °C. For measurements a thermopile was used. The voltages from thermopile caused by the thermionic radiation emitted from the cube are shown in Figure 3 and Figure 4. The figure shows the difference in emitted thermal radiation depending on surface properties. Most radiation is emitted

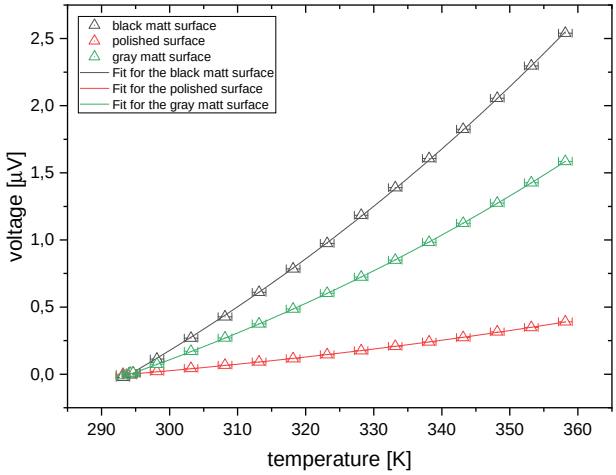

Fig. 3: Data of the Leslie cube measured by a thermopile. The temperature dependence is expected to accord to the Stefan-Boltzmann T^4 law. A fit of the form $A\left(T^B - T_0^B\right)$ was performed for each surface. The data points obtained for the white surface are not shown because they are very close to the pints of the black surface. The errors on the voltage are dropped due to their small size.

by the black matt surface the. Most notably is the similarity with the data obtained for the measured white side. The colour one sees refers to visible wavelengths only. However, the thermopile reacts to thermal radiation which corresponds to a longer wavelength. In this part of the spectrum the white and the black surface have the same "colour" and therefore emit approximately the same amount of radiation. The grey matt surface emits less but shows a similar temperature dependency still. The polished one is the least effective emitter. Since every side is part of the same cube and therefore made out of the same material, the surface texture is responsible for the difference. A matt surface does emit more radiation. The polished one has a high reflectivity. According the Kirchhoff's law the absorbency of a body is also its emissivity. A highly reflective surface has low absorbency and therefore is also a bad emitter. According to the Stefan-Boltzmann law the total energy emitted by a black body is

$$P = \sigma A T^4 \tag{1}$$

where A is the surface's area and $\sigma = 5.670\,\mathrm{Wm^{-2}K^{-4}}$ [1] is the Stefan-Boltzmann constant.

For the data points in Figure 3 a fit of the form $A\left(T^B - T_0^B\right)$ is performed. The reference temperature T_0 was determined to (21.0 ± 0.5)°C. This is the temperature that was set on the thermostat at which the detected voltage observing

the black side was $\approx 0\,\mathrm{V}$. It is important to mention that not the real surface temperature was measured but a temperature at which the cube emits the same amount of thermal radiation as the environment does, so that the thermopile does not detect a temperature gradient. The exponents obtained by the fit are shown in Table 1. The obtained exponents do not coincide with the

side	fit exponent B
black matt	$3,883 \pm 0,057$
polished	$4,23 \pm 0,17$
white matt	$3,781 \pm 0,032$
grey matt	$4,071 \pm 0,041$

Table 1: The exponents determined by the $A\left(T^B - T_0^B\right)$ fit on the data obtained by measurements of the sides of Leslie cube.

expected value of 4, even in range of their error. Nevertheless, the exponents of the black and the white surface are close to 4 but lower. This is not a surprise because the Stefan-Boltzmann law (1) only holds for true black bodies. The Leslie cube is a grey body and therefore a lower exponent is expected. The polished surface shows the biggest deviation. Due to its very low emissivity ambient influences have a bigger impact on the detected radiation flux than on the matt surfaces. Furthermore, the fits of the polished and the grey matt surface only converged after about 7 times more iterations. One error source could be the use of the same coefficient A for the ambient radiation and the one detected from the cube. Also, the temperature dependency of the ambient radiation could be different, resulting in a different exponent. A value above 4 is not plausible because the black body, which is the ideal emitter, has the highest radiant flux. This is similar to the efficiency of a heat engine which cannot exceed the Carnot efficiency. In order to make a direct test whether the Stefan-Boltzmann law holds, Figure 4 shows the data plotted against $T^4 - T_0^4$ to linearise the expected T^4 behaviour. The correlation is highly significant. This plot shows that the Stefan-Boltzmann can be used to describe the data. However, the residues show a bit of a structure which may be caused by a slightly different exponential dependency smaller than 4. This supports the idea that the Leslie cube is a grey body and not a black body.

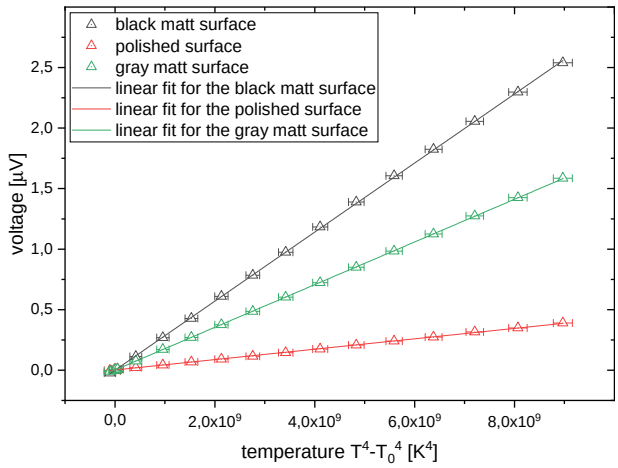

Fig. 4: Data of the Leslie cube plotted against $T^4 - T_0^4$. A linear fit was performed to the data of each surface. The errors on the voltage were dropped due to their small size. The data points for the white surface overlapped the ones for the black surface and therefore were dropped.

2.2 Infrared permeability for several materials

The Leslie cube was oriented with the black side facing the thermopile to achieve a maximum of emitted radiant power. The temperature was set to 75 °C on the thermostat. The voltage U_b generated by the radiation of the black surface without any filter in the mounting is used as reference to determine the transmissivity of different filters. A nickel plate was placed in the filter mounting to determine the offset of $(0.004 \pm 0.010)\mu\mathrm{V}$. A 0.5 mm thick p-doped silicon plate, a 5 mm thick NaCl crystal, an infrared filter and a 3 mm thick window glass were placed in the filter mounting. Then, the generated voltage was recorded. The transmission coefficient for each filter is defined by

$$\tau = \frac{U - U_0}{U_b - U_0} \tag{2}$$

and shown in Figure 5 and Table 2. One can

Table 2: Add caption

	transmissivity [%]	error [%]
window glas	3.71	0.77
infrared filter	2.34	0.77
silicon	47.29	0.78
NaCl crystal	77.50	0.78

see that the infrared filter is the most effective filter. Nevertheless, the observed window glass transmits only 3.71 %,too. Since glass is a non-crystalline, amorphous solid mainly made out of

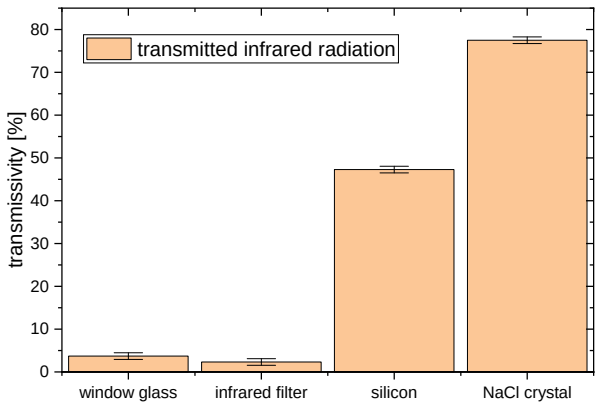

Fig. 5: The infrared transmissivity of different filter materials

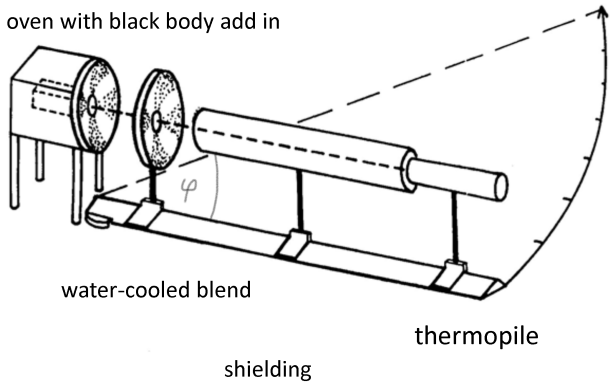

Fig. 6: Experimental setup for measurements on a black body [2] (edited). The blends are water cooled in order to avoid additional radiation. The shielding is also used for blocking additional radiation

SiO_2, the molecules can easily be excited into oscillation by infrared radiation and therefore absorb much of it. The NaCl filter transmits most of the incoming radiation. Due to it's crystalline structure the molecules aren't as easily set into oscillation like in amorphous materials. According to Wien's displacement law

$$\lambda_{peak} = \frac{2.897 \times 10^{-3}\,\mathrm{mK}^{-1}}{T} \tag{3}$$

the maximum intensity of radiation is here at about $\lambda_{peak} = 8.32\,\mu\mathrm{m}$. This corresponds to a energy of 0.15 eV. Silicon has a band gap of 1.12 eV [3] which means it should be completely transparent for the infrared radiation. Nevertheless an absorption of over 50 % is observed. This is due to the p-doping of the silicon plate. Electrons of the valence band can be excited into acceptor levels by a much lower energy that would be needed to excite the electron in the conduction band. The silicon filter therefore is able to absorb infrared radiation which is in accordance with it's transmissivity.

2.3 Black body radiation

A black body is an idealized object with an emissivity of $\epsilon = 1$ and according to Kirchhoff also an absorption coefficient of $\alpha = 1$. The best way to approximate this is by a cavity with a small hole. The cavity is kept at a specific temperature and the emission spectrum of the hole should obey Planck's law.

2.3.1 Measurements on a black body approximation

The black body object was observed at an angle of $\varphi = 0°$ and temperatures between 25 °C and 360 °C. It consisted of an oven that was powered via a variable transformer. The voltage of the transformer was gradually increased to 255 V and then kept at constant. This caused the temperature to rise steadily. During that process the temperature and the voltage on the thermopile were recorded simultaneously. This experimental set up was used to realize a black body as good as possible. The obtained data were again fitted to a form of $A\left(T^B - T_0^B\right)$. An exponent of $B = 3,997 \pm 0,055$ was obtained. This confirms the conjecture that the the Stefan-Boltzmann law can be used to describe the experimental set up. Accordingly, it was shown that the experiment viably realizes a black body.
Again, this is shown in Figure 8, when the radiant flux was observed against the quadruped scale. A linear fit to the scaled data had a correlation coefficient of $r = 0,99834$ making it highly significant that the present data follows the Stefan-Boltzmann law. Still, it can be seen that Stefan-Boltzmann law gives far better values for higher temperatures than for low ones. This can be seen in both 7 and Figure 8. All voltages taken were corrected by the offset voltage that was determined in the first part of the experiment, which especially affected the smaller values. Another reason for the drift in the lower part of the data might be that the black body temperature was close to the working temperature in the laboratory, so that perturbation caused by e.g. the

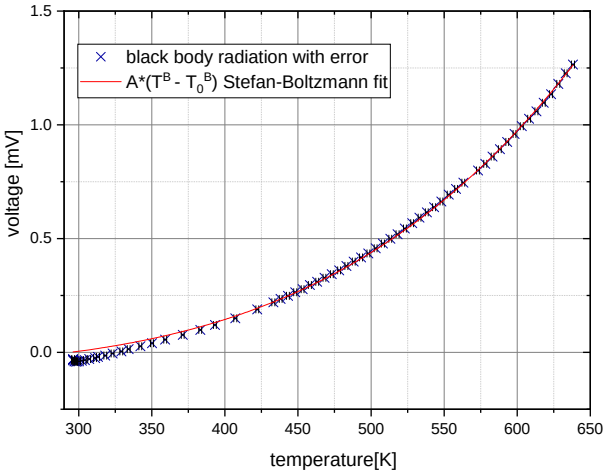

Fig. 7: A black body's radiant flux measured with a thermopile while the temperature was increased; the voltage detected by the thermopile is proportional to the incoming radiant flux and plotted against the rising temperature. One can conjecture the T^4-dependency.

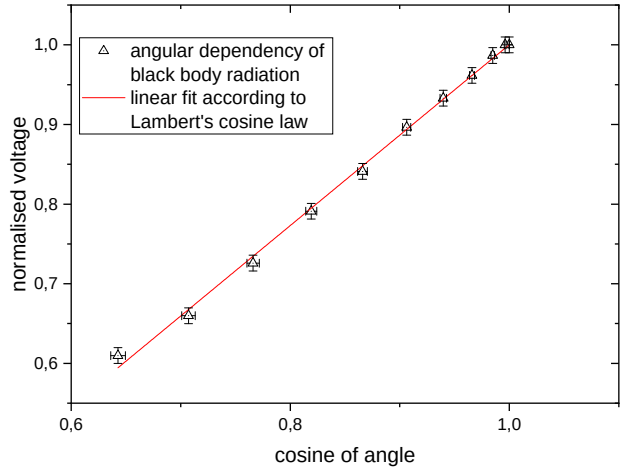

Fig. 9: Angular dependency of the measured radiant flux emitted by a black body. The voltage on a thermopile which is proportional to the incoming radiant flux is normalised and plotted against the cosine of the angle ϕ (see Figure 6). The linear behaviour observed is according to Lambert's cosine law (4)

experimenters could have affected the measurement easily. At higher working temperatures it is far less likely that the body temperature of the experimenters influenced a measurement several hundred Kelvin above it (37 °C). The usage a law describing black body radiation can be seen e.g. in taking the body temperature inside the ear or in calibrating lasers. In astronomy the black body radiation is used to determine the temperature of stars.

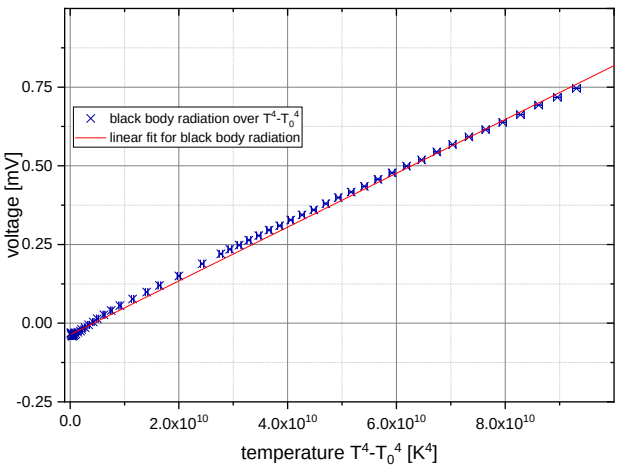

Fig. 8: The same connectivity as in 7. This time it was graphed over the T^4-scale. The little deviation of the linear fit of the data confirms T^4-dependency of the radiant flux to its temperature.

2.3.2 Lambert's cosine law

The luminous intensity of an ideal diffusely reflecting surface or an ideal diffuse radiator is described by Lambert's cosine law

$$J = J_0 \cos \phi \tag{4}$$

This formalism is tested experimentally with the setup shown in subsection 2.3. The oven was heated up to a constant temperature of about 348 °C, ensuring steady radiation. Then, the angle ϕ between oven and thermopile was increased (see Figure 6) and the change of the measured voltage was recorded. The value recorded at 0° was used to normalise all other measured values. The structure in the residues (see Figure 10) may originate from temperature fluctuations. Since there was no temperature control circuit the oven temperature had to be stabilized by choosing a specific voltage on the variable transformer. This however was a hard task because the oven was inert to changes on the transformer. The data of Figure 9 is therefore expected to be slightly influenced by the effect shown by Figure 8.

3 Summary

In this experiment the different realizations of a black body and their functional characterization were probed. The Leslie cube was exemplary for the different stages from a grey to a black body and showed that each surface has its own radiant flux characteristic. The grey matt side turned out to be the best realization of a black body. Only the oven with the hollow cylinder inside

later in the experiment represented the Stefan Boltzmann law more precisely. Additionally, different materials were tested on their transmissivity. It was shown that an infrared coating has the best shielding capacity for thermal radiation. The experiment was topped off in confirming the Lambert law about the angular dependency of thermal radiation.

4 Appendix

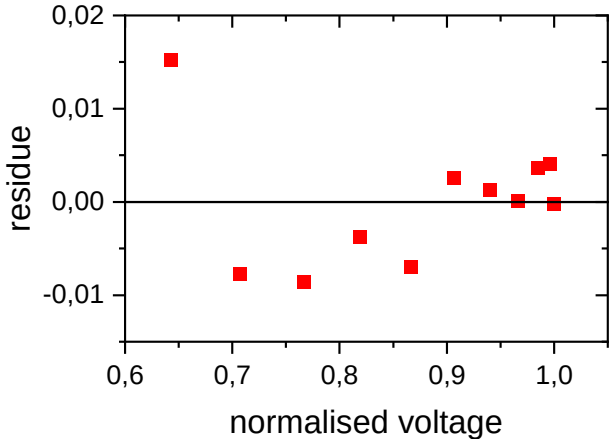

Fig. 10: The residues of the linear fit in Figure 9

References

[1] https://physics.nist.gov/cgi-bin/cuu/Value?sigma last visited September 15th, 2020

[2] Manual for the C33 experiment of the basic internship, Institute of Physics of the University of Würzburg, https://www.physik.uni-wuerzburg.de/studium/bachelor/grundpraktikum/modulbeschreibungen/fortgeschrittenenmodul-c1/, last visited September 15th,2020

[3] https://de.wikipedia.org/wiki/Bandlücke, last visited September 17th,2020